

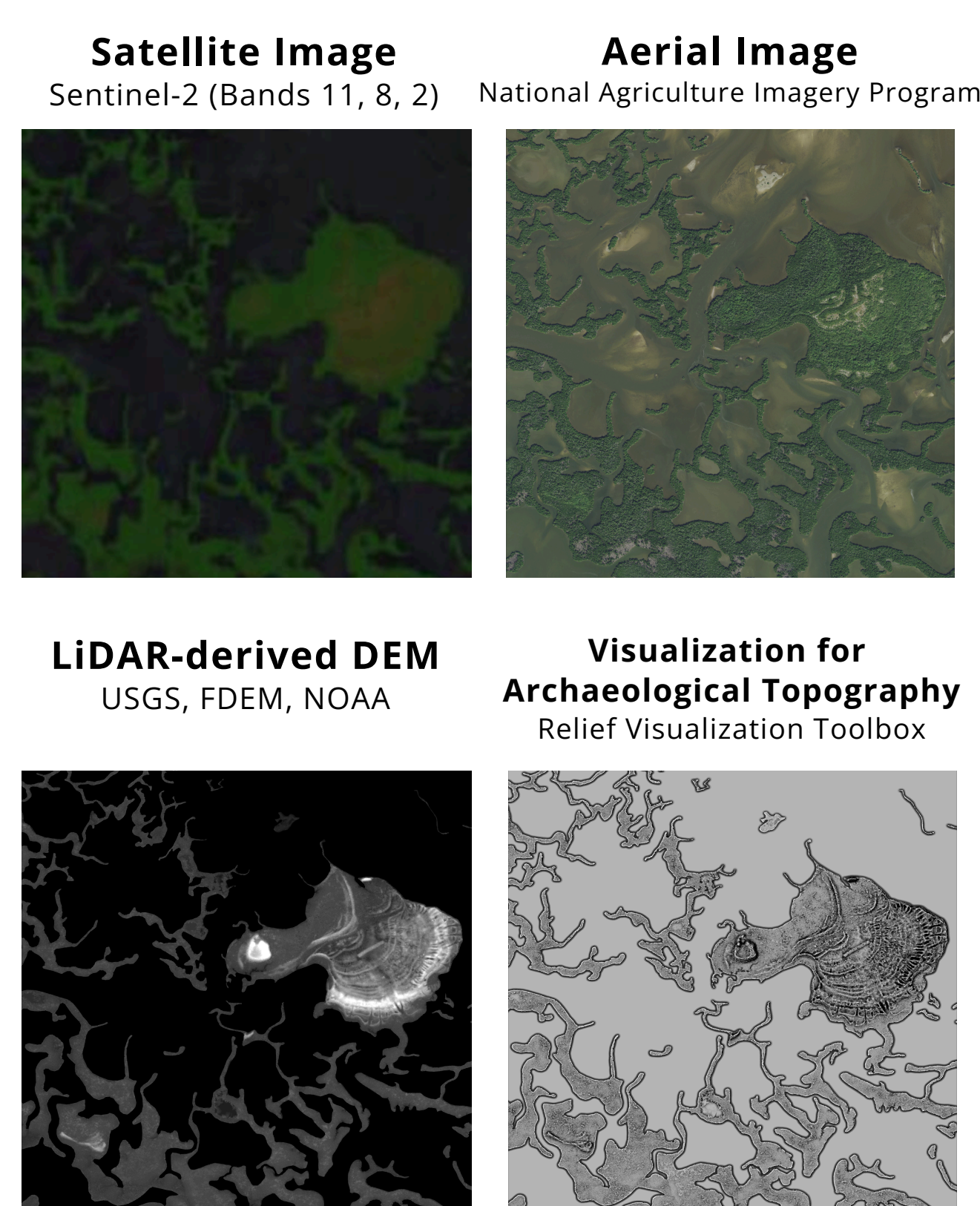
INTRODUCTION

- Shell mounds are large earthworks of shells (clams, oysters, whelks), and other objects, found globally and prevalent along Florida's Gulf Coast.
- Raised fields are terraformed structures created to avoid the destruction caused by intense flooding during the Bolivian wet season.
- These sites provide a unique look at how communities from thousands of years ago lived and altered their natural environments.
- We do not know the locations of many of these sites or whether they have since been destroyed.
- Object detection models can process hundreds of images of Earth's landscape and find underlying patterns.

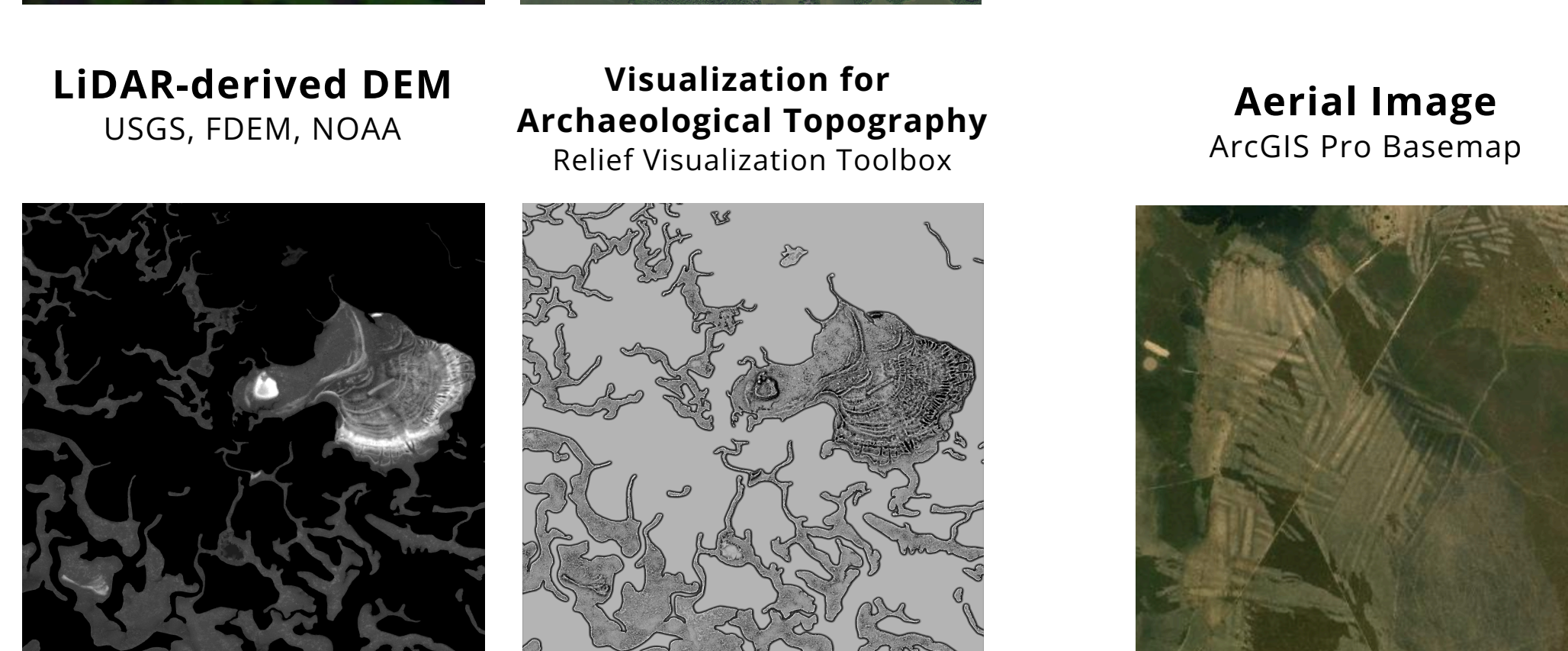
DATA

- Shell mounds
 - Labels from the Florida Master Site File
 - LiDAR-derived digital elevation
 - Sentinel-2 satellite imagery
 - National Agriculture Imagery Program
 - Relief Visualization Toolbox processing
- Raised fields
 - Labels from Dr. John Walker
 - ArcGIS Basemap satellite/aerial imagery

Shell Mounds

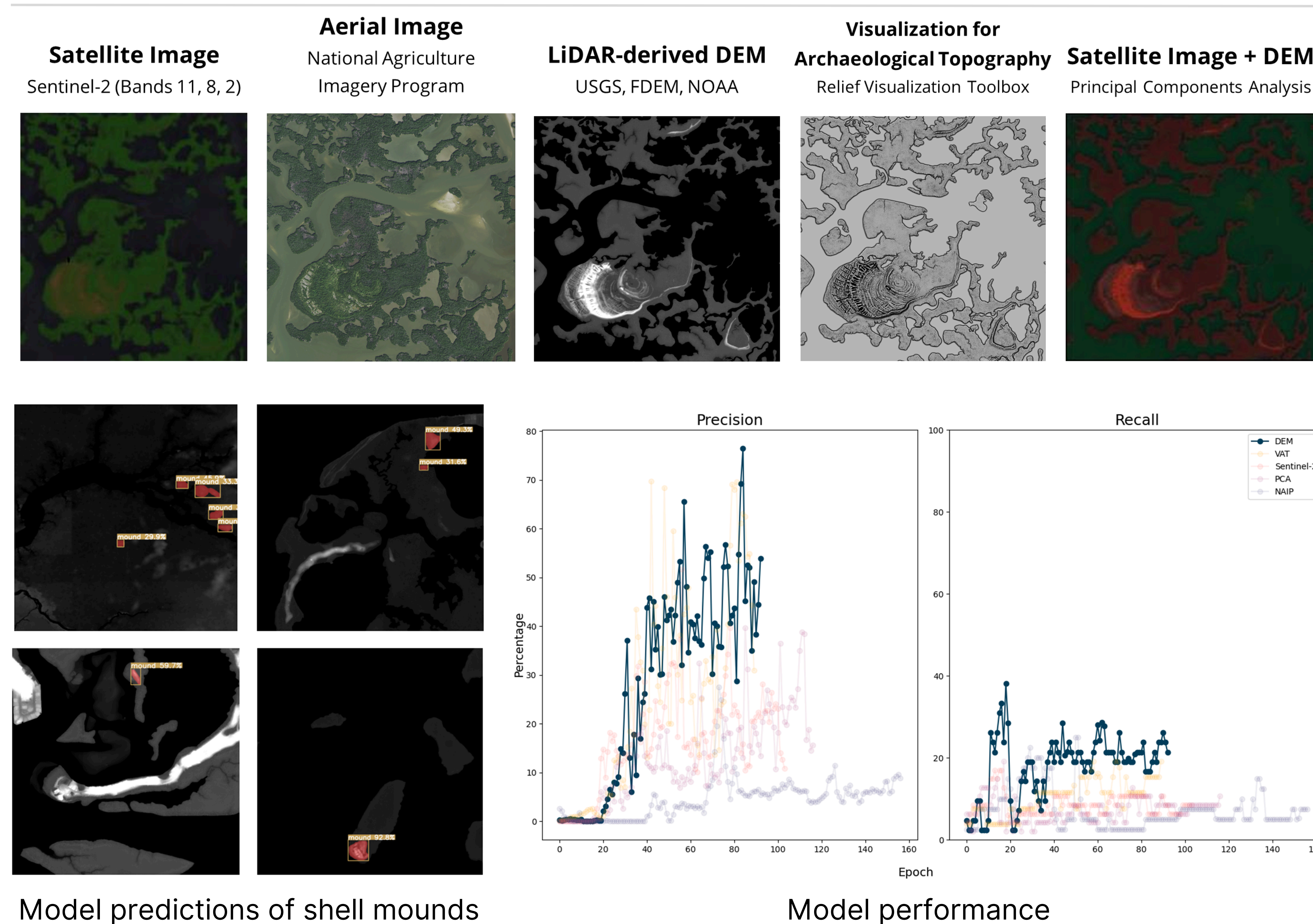


Raised Fields

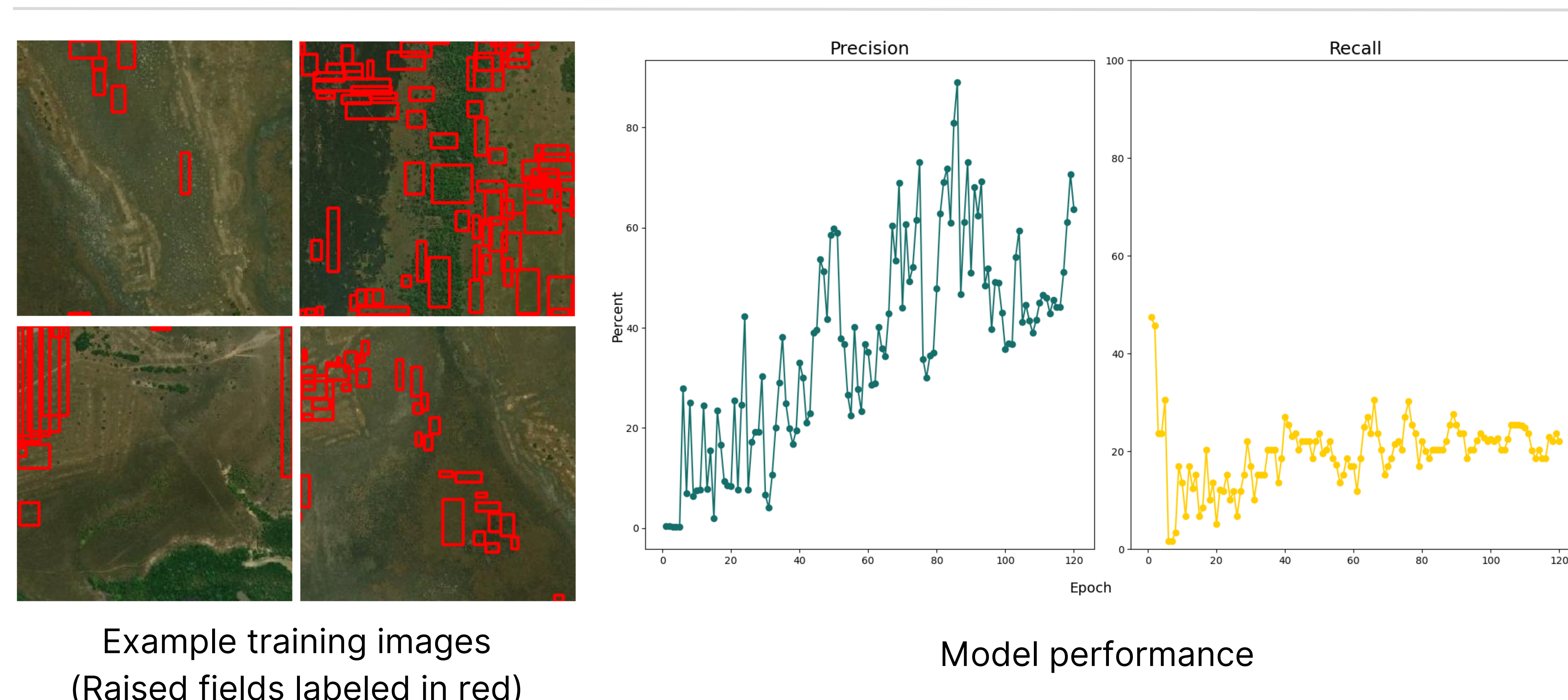


RESULTS

Shell Mounds



Raised Fields



METHODS

- We train an open-source computer vision model, YOLOv8 developed by Ultralytics, in Python to identify archaeological features:
 - Shell mounds in Florida (segmentation)
 - Raised agricultural fields in Bolivia (detection)
- We trained separate models for each data type and archaeological feature, running the models using the University of Florida's supercomputer, the HiPerGator.

DISCUSSION

- Accurately locating archaeological sites is crucial for preserving and protecting these sites.
- One challenge to training the models is the initial lack of data: many potential shell mounds and raised fields may be present *but unlabeled* in the images.
- The trained model has high precision but low recall (of the sites that the model does detect, it is fairly accurate).
- AI and ML can greatly enhance documentation of archaeological sites, but only if applied under the supervision of subject experts and high-quality data.

FUTURE WORK

- Test other models such as Mask-RCNN and YOLOv11.
- Use multi-dimensional remote sensing data to capture unique characteristics.
- Use the predicted outlines to calculate statistics such as change in vegetation and land use/land cover over time.

ACKNOWLEDGEMENTS

- Florida Museum of Natural History Undergraduate Internship Program
- National Science Foundation (BCS-2304807)
- University of Florida Biodiversity Institute Seed Grant, 2024